

Day 111: Regression Evaluation and Cross-Validation Basics

1. Day number

Day 111

2. Topic name

MAE, MSE, RMSE, and Cross-Validation

3. Connection

Yesterday, you evaluated **classification models** using:

Accuracy
Precision
Recall
F1 Score

Those metrics work when the model predicts categories such as:

Spam / Not Spam
Churn / No Churn

Today, we return to **regression**, where the model predicts numerical values such as:

House price → ₹52 lakh
Temperature → 29°C
Sales → ₹85,000

Now we need to measure:

| **How far are the predicted numbers from the real numbers?**

We will use:

MAE
MSE
RMSE

Then we will learn **cross-validation**, which gives us a more reliable way to evaluate a model using different portions of the available data.

4. Important topics

Prediction error

For one example, prediction error is the difference between:

Actual value
and
Predicted value

Suppose:

Actual house price = 50 lakh
Predicted house price = 47 lakh

The model missed by:

3 lakh

Regression evaluation metrics summarize errors like this across many predictions.

MAE

MAE = Mean Absolute Error

Think:

| On average, how far away are my predictions from the actual values?

MSE

MSE = Mean Squared Error

Think:

| Square each error before averaging, so large mistakes receive much more attention.

RMSE

RMSE = Root Mean Squared Error

Think:

| Start with MSE, then take the square root so the result returns to the original unit.

Fold

A **fold** is one portion of a dataset used during cross-validation.

For example, with 5-fold cross-validation:

Dataset
↓
Fold 1
Fold 2
Fold 3
Fold 4
Fold 5

Cross-validation

Cross-validation repeatedly changes which fold is used for validation.

This helps us see whether the model performs reasonably across different parts of the dataset.

5. Foundational notes

Suppose a house-price model makes these predictions:

House	Actual price	Predicted price
A	40	38
B	50	54
C	60	59

The errors in size are:

House A $\rightarrow$ 2 House B $\rightarrow$ 4 House C $\rightarrow$ 1

A perfect prediction would have:

Error = 0

Generally, for MAE, MSE, and RMSE:

 | **Smaller is better.**

And:

0 = perfect predictions

However, what counts as a "small" error depends on the meaning and scale of your target.

An RMSE of 5 could be tiny for house prices measured in lakhs but enormous for something measured between 0 and 1.

6. MAE, MSE, and RMSE intuitively

MAE — average size of mistakes

Imagine your prediction errors are:

2
4
1

MAE simply asks:

| What is their average size?

It does not care whether the model predicted too high or too low.

That's why we use **absolute** values.

Simple formula:

MAE =
sum of absolute errors

number of predictions

For our errors:

MAE = (2 + 4 + 1) / 3

You can calculate the final value yourself.

How to interpret it

If:

MAE = 3 lakh

you can say:

| The model's predictions are off by about 3 lakh on average.

This direct interpretation is one reason MAE is beginner-friendly.

7. MSE — punish large mistakes more

MSE first **squares each error**.

Suppose two errors are:

2
10

Squaring gives:

$2^2 = 4$
 $10^2 = 100$

Notice how the large error becomes much more influential.

Simple formula:

$$\text{MSE} = \frac{\text{sum of squared errors}}{\text{number of predictions}}$$

So MSE is useful when you want large prediction mistakes to receive extra emphasis.

One drawback is that its units are squared.

If the target is measured in:

lakh rupees

MSE is effectively measured in:

(lakh rupees)²

which is less intuitive to explain directly.

8. RMSE — MSE returned to the original unit

RMSE means:

Root Mean Squared Error

You first calculate MSE and then take its square root:

$$\text{RMSE} = \sqrt{\text{MSE}}$$

The main benefit is that RMSE returns to the same unit as the target.

If house prices are measured in lakhs:

RMSE → lakhs

rather than squared lakhs.

So you can interpret something like:

$$\text{RMSE} = 4 \text{ lakh}$$

more naturally.

A useful comparison:

Metric	Main intuition	Large errors
MAE	Average absolute mistake	Normal influence
MSE	Average squared mistake	Strong influence
RMSE	Square-root version of MSE	Strong influence

9. Easy example

Suppose a model predicts monthly sales.

Month	Actual	Predicted
1	10	12
2	20	18
3	30	33
4	40	39

First find the errors:

Actual 10, Predicted 12 → difference = 2
 Actual 20, Predicted 18 → difference = 2
 Actual 30, Predicted 33 → difference = 3
 Actual 40, Predicted 39 → difference = 1

For MAE, you would average:

2, 2, 3, 1

For MSE, square them first:

2^2 , 2^2 , 3^2 , 1^2

For RMSE:

calculate MSE
 ↓
 take its square root

Do the final arithmetic yourself as today's exercise.

10. Cross-validation — dataset rotation analogy

Suppose you have five groups of data:

A B C D E

Instead of choosing one validation group forever, cross-validation **rotates** the validation role.

Round 1

Train: B C D E
Validation: A

Round 2

Train: A C D E
Validation: B

Round 3

Train: A B D E
Validation: C

Continue until every group has had a turn as validation data.

This is similar to a group of students taking turns being the examiner:

Round 1 → Student A checks everyone else
Round 2 → Student B checks everyone else
Round 3 → Student C checks everyone else
...

No single group permanently decides the evaluation result.

11. Why use cross-validation?

Imagine you perform only one train/validation split.

By chance, your validation data might be:

very easy

or:

unusually difficult

Your evaluation could therefore give a misleading impression.

Cross-validation asks:

| Does the model behave reasonably when different parts of the dataset are held out?

For example:

Fold 1 MAE $\rightarrow$ 3.1
Fold 2 MAE $\rightarrow$ 2.8
Fold 3 MAE $\rightarrow$ 3.4
Fold 4 MAE $\rightarrow$ 2.9
Fold 5 MAE $\rightarrow$ 3.2

You could then summarize the results using their average.

Conceptually:

Train/evaluate several times
 $\downarrow$
collect several scores
 $\downarrow$
look at average performance

This gives a broader picture than relying on one lucky or unlucky split.

12. Problem statement

Suppose these are actual and predicted house prices, measured in lakhs:

House	Actual	Predicted
A	30	28
B	40	43
C	50	49
D	60	64

Part A — Regression errors

For each house:

1. Find the prediction error size.
2. Use the absolute errors to calculate MAE.
3. Square the errors and calculate MSE.
4. Take the square root of MSE to obtain RMSE.
5. Explain each result in simple language.

Part B — Cross-validation

Imagine you have:

12 training examples

and use:

3-fold cross-validation

Conceptually divide them into:

Fold A → 4 examples
Fold B → 4 examples
Fold C → 4 examples

Explain how the three rounds would work:

Round 1 → ? used for validation
Round 2 → ? used for validation
Round 3 → ? used for validation

Do not build a complicated model. Focus on understanding the rotation.

13. Concepts used

This lesson combines:

Actual value
Predicted value
Prediction error
Absolute error
Squared error
MAE
MSE
RMSE
Training data
Validation data
Fold
Cross-validation

The important distinction is:

Error metrics
→ measure how wrong predictions are

Cross-validation
→ changes which examples are used to evaluate the model

They solve related but different problems.

14. Thought process

For regression evaluation, start with one row.

Suppose:

Actual = 30
Predicted = 28

Ask:

How far apart are they?

Then repeat for every prediction.

Once you have all errors:

MAE
→ make errors positive and average

MSE
→ square errors and average

RMSE
→ square root of MSE

For cross-validation, think differently.

Ask:

How can I avoid judging my model
using only one particular validation group?

Answer:

Rotate the validation group.

So the full conceptual picture becomes:

```
Train model
  ↓
make predictions
  ↓
calculate errors
  ↓
repeat evaluation with different folds
  ↓
summarize performance
```

15. Beginner-friendly calculation steps

MAE

Given:

Actual: 30, 40, 50, 60 Predicted: 28, 43, 49, 64

Step 1:

Find each difference

Step 2:

Ignore the + or - direction by taking absolute values
--

Step 3:

Add those absolute errors

Step 4:

Divide by 4

That gives:

MAE

MSE

Start with the same errors.

Then:

square error 1 square error 2 square error 3 square error 4 add them divide by 4

That gives:

MSE

RMSE

Take your MSE:

RMSE = square root of MSE

Cross-validation pseudocode

```
split dataset into k folds
FOR each fold:
    use that fold as validation data
    use all other folds as training data
    train the model
    predict validation examples
    calculate an error score
store the score
calculate the average score
```

At this stage, understanding this workflow matters more than writing complete Python code.

16. Easy edge cases

Perfect prediction

Suppose:

Actual: 10, 20, 30
Predicted: 10, 20, 30

Every error is:

0

Therefore:

MAE = 0
MSE = 0
RMSE = 0

This represents perfect predictions on those examples.

One very large error

Suppose the errors are:

1
2
1
20

MAE sees the large error, but MSE squares it:

$$20^2 = 400$$

That one mistake can therefore influence MSE and RMSE strongly.

This demonstrates an important difference:

| MSE and RMSE react more strongly to large errors than MAE.

Very small dataset

Suppose you have only three examples.

Trying to create many folds would leave very little training data in each round.

Cross-validation can still be conceptually possible, but evaluation from extremely small datasets may be unstable.

More folds are not automatically better in every situation.

Unequal-looking fold performance

Imagine:

Fold 1 MAE = 2
Fold 2 MAE = 3
Fold 3 MAE = 15

The model behaves very differently on Fold 3.

Do not look only at the average and ignore this variation.

At beginner level, simply notice:

Cross-validation can reveal that performance changes depending on which data the model is evaluated on.

17. Expected interpretation

After solving the regression exercise, your explanation should sound like:

MAE:

Predictions are off by about ____ lakh on average.

MSE:

The average squared error is ____.
Larger mistakes receive extra weight.

RMSE:

The typical error measured in the original target unit is about ____ lakh.

For cross-validation:

Each fold gets one turn as validation data.

The remaining folds are used for training.

Therefore, the model is evaluated on multiple different portions of the dataset.

You should understand the meaning of the numbers rather than simply calculating them.

18. Hint only

For the house-price problem, start here:

Actual	Predicted	Absolute error
30	28	?
40	43	?
50	49	?
60	64	?

Then:

MAE

→ average the absolute-error column

MSE

→ square each error, then average

RMSE

→ $\sqrt{\text{MSE}}$

For the 3-fold exercise:

Round 1:

Train = Fold ? + Fold ?

Validate = Fold ?

Round 2:

Train = Fold ? + Fold ?

Validate = Fold ?

Round 3:

Train = Fold ? + Fold ?

Validate = Fold ?

The key idea from **Day 111** is:

MAE, MSE, and RMSE tell us how far numerical predictions are from actual values, while cross-validation evaluates the model repeatedly by rotating which portion of the data is held out for validation.